Limbayong

Deepwater Oil & Gas Development · Blocks G / J / R · Offshore Sabah

Review Pack v0.1 — Parts 1 and 3, with ledger and source register

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Revision	0.1 — first review issue
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Prepared by	HERMES (arifOS), for review
Status	Draft for review — not for external circulation
Scope	Parts 1 and 3, Appendix B, Appendix E
Classification	Independent reconstruction from public sources

READ THIS FIRST

This document is not a PETRONAS or PETRONAS Carigali document. It was not produced with, reviewed by, or endorsed by either company. It contains no internal data, no well logs, no seismic, no PVT, no cost estimates and no fiscal terms from any operator. Every number is either taken from a cited public source, derived by arithmetic from those sources, or flagged as an assumption with its owner and reason. Where a figure was not available, the gap is marked OPEN rather than filled.

Revision 0.1 builds only the parts that do not depend on the Sevan-300 specification or the fiscal reconciliation. Parts 0, 2 and 4–12 are deliberately absent. Their numbers would inherit the same defects this pack exists to remove.

Document control

Revision block

Rev	Date	Change	Scope
0.1	19 Oct 2026	First review issue	Front matter, Part 1, Part 3, App. B, App. E

Reading guide — the four tags

Tag	Meaning and test
VERIFIED	Stated by a cited public source. The citation is in Appendix E and the row is in Appendix B. If the source is wrong, the number is wrong — but the document is faithful to it.
DERIVED	Computed here from other ledger rows. The formula is shown in the row, so the arithmetic can be repeated without trusting the author.
ASSUMED	A judgement, not a measurement. Every one carries an owner and a stated reason. An assumption can be challenged; it cannot be verified.
OPEN	The input does not exist in the public record. It is marked, not guessed. OPEN items appear as red banners wherever they change a conclusion.

The rule this pack is built on: no number may exist only in prose. Every figure in Parts 1 and 3 carries a ledger row reference (L-nn) in Appendix B. If a number is found without a ledger row, that is a defect in this document, not a rounding matter.

Data classification statement

All information in this pack derives from public sources: a company filing on Bursa Malaysia, published petroleum-engineering papers, trade press, an open-source geospatial dataset, and a public resource tracker. No PETRONAS, PETRONAS Carigali, MISC or contractor confidential material has been used, requested, or received.

Nothing in this pack should be read as reflecting internal operator knowledge. Any reader with access to operator data should expect this document to differ from it, and should prefer the operator’s data.

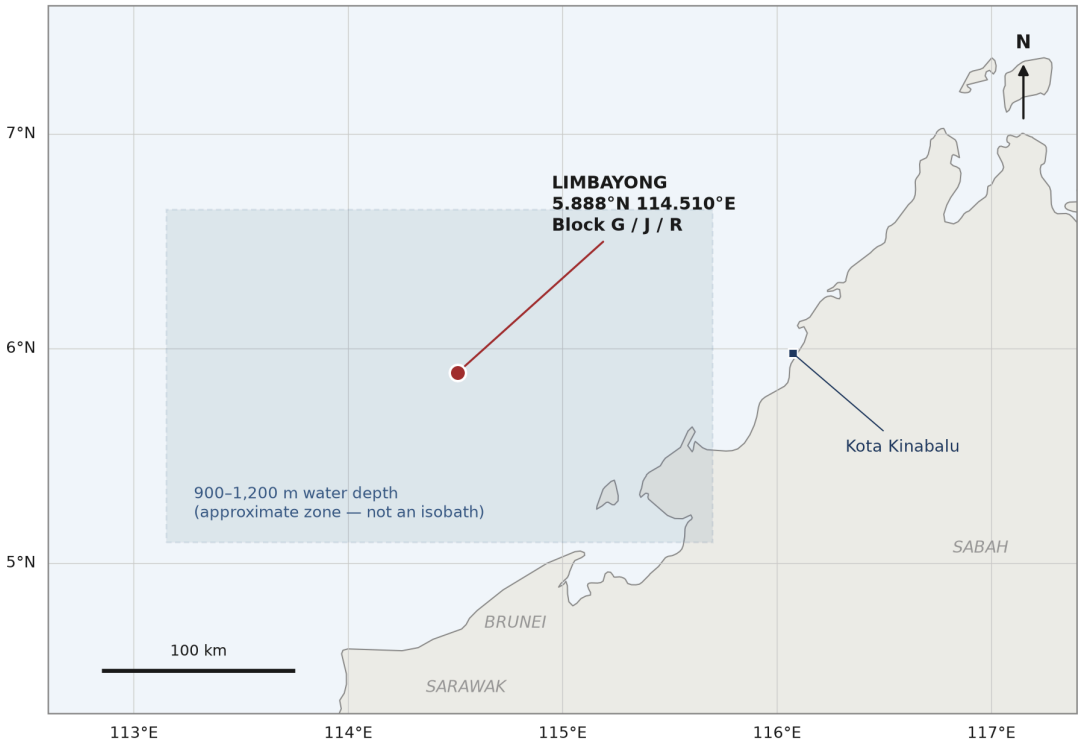
Not in this build

Parts 0, 2 and 4–12 are not drafted. Part 2 and Part 4 both consume the fiscal reconciliation, which does not yet exist as a single engine; Part 4 additionally consumes the Sevan-300 specification, which has not been established from a primary source. Drafting them now would repeat the exact failure this pack was commissioned to fix — many workstreams, many number sets.

1 Basis and reconciliation

1.1 Basis of design

For any reader outside the operator, Limbayong presents a specific problem: the public record contains enough to know that a development exists, and not enough to know whether it works. This pack therefore separates three things that are usually blended together.



Source: Natural Earth 10m coastline; position from Global Energy Monitor Global Oil & Gas Extraction Tracker (data year 2023).
Depth zone is schematic — no public isobath used. Block outline not public; point only. Projection WGS84.

Figure 0 — Location. Limbayong lies in Blocks G/J/R, approximately 120 km offshore Sabah in 900–1,200 m water depth. The shaded zone is schematic and is not an isobath; the block outline is not public, so a point is shown rather than a polygon. Base map: Natural Earth 1:10m coastline. Position: L-17. VERIFIED

What is established. That the field lies in Blocks G/J/R about 120 km offshore Sabah in 900–1,200 m of water, that it was discovered in 2002, that development is a ten-well subsea tie-back to a floating production unit, that final investment decision was taken on 28 August 2026, and that the working interest is 100 per cent PETRONAS Carigali. These are cited facts. VERIFIED

What is inferred. That a structure 30 km long and 2.5 km wide, described by its own operator as “industry top decile reservoir complexity” with “lateral compartmentalisation”, cannot deliver a uniform recovery factor across its length. The inference is sound, but it is an inference. DERIVED

What is absent. There is no public seismic, no public well log, no core, no PVT sample, no production test, no cost estimate, no gas sales agreement and no fiscal term sheet. Not “not found by this author” — absent from the public record. Every subsurface volume in this pack is therefore a screening range, not a reserves estimate.

Language note. This pack uses SPE-PRMS vocabulary. Nothing here is called “reserves”. The volumes in Part 3 are **contingent resources under a development case**, expressed as a range because the basis for a point estimate is not public.

1.2 One number set

The table below is the only accepted source of values for Parts 1 and 3. Every number in those parts traces to a row here, and every row traces to Appendix B.

Parameter	Value used	Class	Row	Basis
Oil in place — low	347 MMbbl	ASSUMED	L-01	Screening range
Oil in place — mid	372.5 MMbbl	DERIVED	L-02	(347 + 398) / 2
Oil in place — high	398 MMbbl	ASSUMED	L-03	Screening range
Recovery factor — low	35 %	ASSUMED	L-04	Screening range
Recovery factor — mid	37.5 %	DERIVED	L-05	(35 + 40) / 2
Recovery factor — high	40 %	ASSUMED	L-06	Screening range
Recoverable — low	121.5 MMbbl	DERIVED	L-07	347 × 0.35
Recoverable — mid	139.7 MMbbl	DERIVED	L-08	372.5 × 0.375
Recoverable — high	159.2 MMbbl	DERIVED	L-09	398 × 0.40
Plateau oil — low	30,000 bopd	ASSUMED	L-10	Screening range
Plateau oil — mid	33,507 bopd	DERIVED	L-11	(30,000 + 37,014) / 2
Plateau oil — high	37,014 bopd	ASSUMED	L-12	Screening range
Gas recoverable	790.9 bcf	VERIFIED	L-13	Public tracker, data year 2023
Gas nameplate	150 MMscfd	VERIFIED	L-14	Operator release + tracker
Oil nameplate	40,000 bopd	VERIFIED	L-15	Operator release, 2021
Water depth	900–1,200 m	VERIFIED	L-16	Operator release, 2021
Working interest	100 % PCSB	VERIFIED	L-18	Trade press, 2026
First oil	—	OPEN	L-19	Not determinable in this pack
Economic limit	—	OPEN	L-20	Requires the fiscal engine
Injection duty	—	OPEN	L-21	No public basis
Fiscal terms	—	OPEN	L-25	PSC text not public

Table 1.2 — One number set for Parts 1 and 3. Values are quoted from ledger rows, not restated from prose. Class tags are defined in the reading guide.

1.3 Conflicts — resolved and open

This pack assembles nine technical sections produced in parallel. They did not agree. Seven disagreements were identified. Two are resolved by definition; five remain open and are carried forward to the parts that must close them.

Every open conflict appears as a red banner in the body of the affected section, and is listed once here. No conflict has been resolved silently.

Why C-02 is resolved rather than open. The apparent conflict between 40,000 bopd and 35,000 bopd is not a disagreement about the field. It is a category error: 40,000 is the facility's nameplate capacity, and 35,000 was a forecast of what the reservoir could deliver into it. Once the two are named separately there is no conflict — there is a shortfall, which is a finding. The same applies to gas at 150 and 135 MMscfd.

ID	Conflict	Values in the record	Status	Owner	Next action
C-01	Total development capex	2.83 / 2.90 / 3.17 / 4.05 bn	OPEN	Part 4	Independent cost build; adopt one WBS, retire the other three
C-02	Throughput: nameplate or forecast	40,000 vs 35,000 bopd; 150 vs 135 MMscfd	RESOLVED	Part 1	Separate the two objects; Part 3 uses 30,000–37,014 bopd and 150 MMscfd nameplate
C-03	Oil in place	517 MMbbl STOIIP vs 347–398 MMbbl	OPEN	Part 5	STOIIP and screening OOIP are different definitions; adopt one basis
C-04	Net present value at one price	4 values at one price	OPEN	Part 4	Single fiscal engine; no NPV is quoted in this pack
C-05	First oil	2029 vs Q4 2030	OPEN	Part 6	Integrated schedule against comparable projects
C-06	Decline requirement	"44 %" stated	OPEN	Part 3	Not reproducible from the stated ranges — see §3.2 and Figure 2
C-07	Water injection duty	46,300 vs 75,000 bwpd	OPEN	Part 7	Required voidage is derivable from rate and FVF; stated duty is not

Table 1.3 — Conflict register. Seven conflicts identified across the parallel sections. Two resolved by definition, five open and carried forward.

1.4 When a range is a range

A range is not a hedge. Three ranges in this pack are load-bearing and each is held open for a different reason.

Oil in place, 347–398 MMbbl
Open because no public OOIP exists. A screening bracket, not a probabilistic one. ASSUMED

Recovery factor, 35–40 %
Open because the recovery mechanism — solution gas, gas cap, water injection — is not stated publicly. ASSUMED

Plateau, 30,000–37,014 bopd
Open because well count is stated but well productivity is not. ASSUMED

3 Production

3.1 The three cases

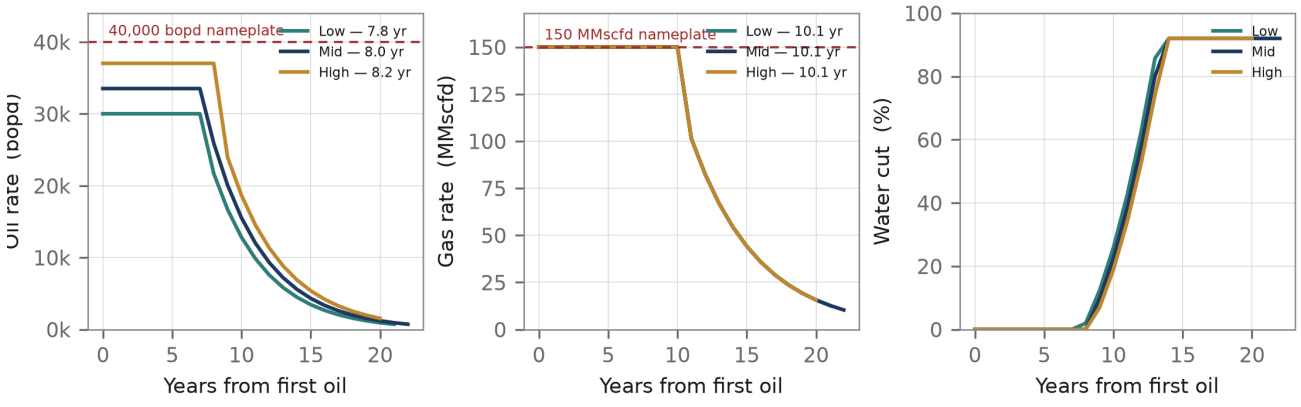
Three numbers govern the production case, and they cannot be chosen independently.

- Recoverable volume — 121.5 to 159.2 MMbbl DERIVED (L-07 to L-09)
- Plateau rate — 30,000 to 37,014 bopd ASSUMED (L-10, L-12)
- Field life — the dependent variable, not a choice

They are run as three cases rather than a point estimate, because the basis for a point estimate does not exist in the public record.

	Low	Mid	High
Oil in place	347 MMbbl	372.5 MMbbl	398 MMbbl
Recovery factor	35 %	37.5 %	40 %
Recoverable	121.5 MMbbl	139.7 MMbbl	159.2 MMbbl
Plateau rate	30,000 bopd	33,507 bopd	37,014 bopd
Plateau duration	7.8 yr	8.0 yr	8.2 yr
Gas plateau	10.1 yr	10.1 yr	10.1 yr
Plateau share of ultimate	70 %	70 %	70 %
Decline	exponential	exponential	exponential
Economic limit	OPEN	OPEN	OPEN

Table 3.1 — Case definition. Gas plateau duration is identical across cases because the gas resource is one ledger value (L-13); only oil volumes vary. The economic limit is deliberately not stated — it is OPEN (L-20), see §3.5.



Cases: Low 347 MMbbl / 35% / 30,000 bopd · Mid 372.5 / 37.5% / 33,507 · High 398 / 40% / 37,014.
Plateau carries 70% of ultimate (ASSUMED); exponential decline exhausts the rest. Water cut ASSUMED.
Economic limit OPEN — tail truncation not determinable in this pack.

Figure 1 — Production cases. Oil, gas and water cut for Low, Mid and High. The oil plateau sits **below** the 40,000 bopd nameplate in all three cases — that gap is the finding, not a plotting error. The gas stream holds nameplate longer than oil in every case. Sources: L-01 to L-15. DERIVED

The first thing Figure 1 shows is the one that matters: **no case reaches the nameplate.** The facility is designed for 40,000 bopd and the reservoir delivers between 30,000 and 37,014. That is a 7.5 to 25 per cent shortfall against installed capacity, present from first oil and permanent thereafter.

3.2 The plateau collision

At a sustained plateau with no decline, the resource is exhausted in the time it takes to divide recoverable volume by annual plateau volume.

Case	Recoverable	Plateau	Years to exhaustion
Low	121.5 MMbbl	30,000 bopd	**11.1 years**
Mid	139.7 MMbbl	33,507 bopd	**11.4 years**
High	159.2 MMbbl	37,014 bopd	**11.8 years**

The three cases converge on the same answer because numerator and denominator move together. That convergence is the finding: **within this resource range, the plateau rate does not buy additional years. It only changes which years they are.**

3.2.1 The stress case: 9.6 years

The brief for this pack cites a 9.6-year figure. It reproduces exactly — but only under one specific pairing.

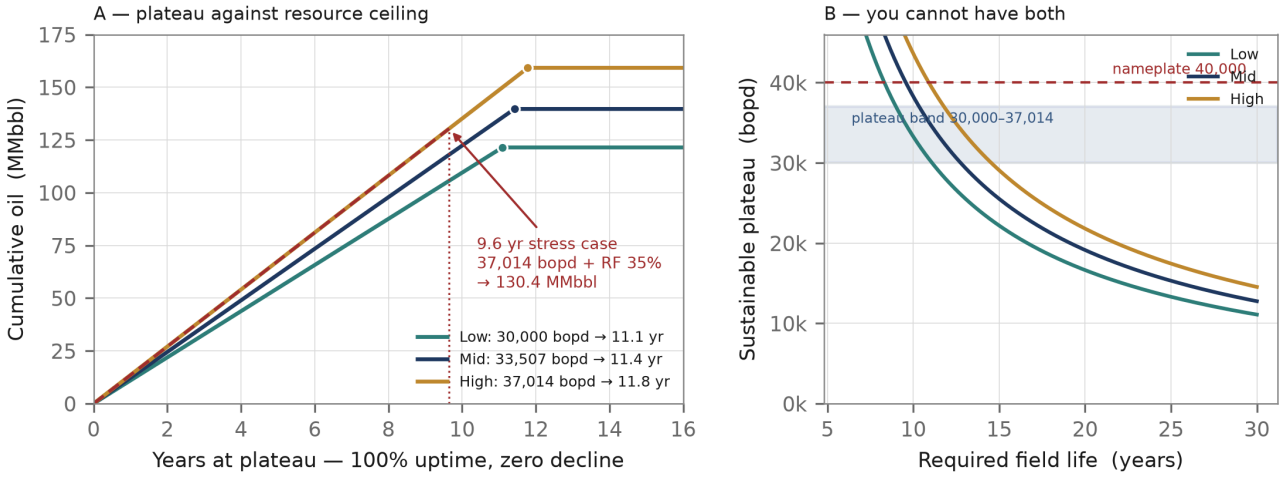
9.6-YEAR ARITHMETIC

372.5 MMbbl (mid OOIP) × 0.35 (low recovery factor) = **130.4 MMbbl**

130.4 MMbbl ÷ (37,014 bopd × 365) = **9.65 years**

The optimistic rate paired with the pessimistic recovery. Neither is a base case; the pairing is the stress.

OPEN — C-06 — the "44 % decline requirement" does not reproduce. If the field is held at 37,014 bopd the resource is exhausted in 9.65 years and there is no tail to decline **into**. Working the other direction — a 20-year field life at the same initial rate — requires an exponential decline of approximately **8.7 % per year**, from `(1 − (1−d)^20) / d = 9.65`. Forty-four per cent appears nowhere in that arithmetic. The figure is therefore carried as OPEN. Either it refers to a quantity not stated in the brief, or it is in error. It is not silently accepted and it is not silently dropped.



DERIVED: plateau years = recoverable ÷ (rate × 365); recoverable = OOIP × RF. Panel A stress case pairs the high rate with the low recovery factor at mid OOIP.
The '44%' decline figure in the build brief does not reproduce from these ranges — see Table 1.3, conflict C-06 (OPEN).

Figure 2 — Plateau against resource ceiling. **Panel A:** cumulative oil at a sustained plateau for each case; markers show where each hits its recoverable volume. The red dashed line is the 9.6-year stress case. **Panel B:** the same resource expressed as required field life. The nameplate is 40,000 bopd; the shaded band is the 30,000–37,014 bopd plateau range.

Sources: L-01 to L-12. DERIVED

The consequence for the facility is direct. Panel B shows that a 20-year field life at this resource range supports a plateau of roughly **17,000 to 22,000 bopd** — between 42 and 55 per cent of the 40,000 bopd nameplate. A development designed for 40,000 bopd and a 20-year life is asking for two things the resource does not offer at once.

3.3 Gas against the 150 MMscfd nameplate

Gas behaves differently, and that difference is the second half of the problem.

- Recoverable gas — 790.9 bcf VERIFIED (L-13)
- Nameplate — 150 MMscfd VERIFIED (L-14)
- Full-rate exhaustion — 790.9 bcf ÷ 54.75 bcf per year = **14.4 years**

At nameplate the gas stream outlasts the oil stream at plateau by roughly five years, but likewise has no tail if drawn at full rate. Applying the same convention used for oil — plateau carries 70 per cent of ultimate recovery — the gas plateau runs **10.1 years** against the oil plateau's 7.8 to 8.2 years.

The collision, stated plainly. For roughly two years the facility would process gas with little or no oil. That tail is long in calendar time and thin in revenue. A floating unit's operating cost does not fall with throughput, so the unit cost of each barrel in that period is fixed cost divided by a small volume. That is a Part 4 question, but it is a Part 3 **fact**, and it should not be discovered for the first time inside an economic model.

OPEN — Not established: whether the 150 MMscfd nameplate is a gas-processing limit, a sales-route limit, or a reservoir-deliverability limit. Those three have completely different consequences. A process limit can be relieved with capital; a sales-route limit needs a buyer; a deliverability limit needs wells. The public record does not say which it is. (L-14)

3.4 Waterflood and voidage

If the field is waterflooded — and the public record neither confirms nor denies it — the injection duty is not an independent design choice. It follows from the production rate and the formation volume factor.

Quantity	At 30,000 bopd	At 37,014 bopd
Reservoir voidage at Bo 1.25	37,500 rb/d	46,268 rb/d
Injection at 100 % replacement	37,500 bwpd	46,268 bwpd
Injection at 80 % replacement	30,000 bwpd	37,014 bwpd

The formation volume factor of 1.25 is ASSUMED (L-22) — no public PVT exists. The 100 per cent replacement case is a design ceiling, not a recommendation.

OPEN — C-07 — required voidage and stated duty do not agree. Derivation gives roughly 46,300 bwpd at the high plateau. A parallel section in this programme states a 75,000 bwpd water-injection requirement. No basis for the higher figure is reproducible from rate and formation volume factor; it may include margin, a different rate assumption, or produced-water disposal. Until the basis is established the injection duty is OPEN, and no facility sizing should be quoted from this pack.

3.5 What these cases are not

They are not forecasts. A forecast requires an economic limit — the rate at which operating cost consumes the margin on the incremental barrel. Setting that rate requires a price deck, an opex structure and a fiscal regime, none of which exist in this pack.

OPEN — L-20 — economic limit. Not stated anywhere in this document. Every profile in Figure 1 therefore runs until the resource is exhausted at an arbitrary 2 per cent of plateau, which is a plotting convenience and not an economic cut-off. The true tail length is unknown, and with it the true field life, the abandonment timing and the late-life unit cost. This is the largest gap in Part 3.

3.6 What Part 3 establishes and what it does not

Established

- No case reaches the 40,000 bopd nameplate. The shortfall is 7.5 to 25 per cent, present from first oil.
- At this resource range the field cannot sustain 40,000 bopd for more than about nine to twelve years, and cannot do so with any decline tail.
- The 9.6-year figure is real and derives from one specific pairing of optimistic rate with pessimistic recovery.
- Gas at nameplate outlasts oil at plateau by roughly two years under a common convention, producing a thin, low-revenue tail.
- Required water-injection duty at the high plateau is of order 46,000 bwpd.

Not established

- Whether 10 wells can deliver 37,014 bopd. No productivity data is public.
- Whether the recovery factor is 35 or 40 per cent.
- Whether water injection is planned at all.
- Where the economic limit sits (OPEN, L-20), and therefore how long the field runs.
- What the gas nameplate constraint actually represents.

Appendix B — Assumptions ledger

Only rows used by Parts 1 and 3 are listed. The full ledger will be issued with the fiscal engine.

Row	Parameter	Value	Class	Owner / basis	Citation
L-01	OOIP — low case	347 MMbbl	ASSUMED	Review brief	—
L-02	OOIP — mid case	372.5 MMbbl	DERIVED	(L-01 + L-03) / 2	—
L-03	OOIP — high case	398 MMbbl	ASSUMED	Review brief	—
L-04	Recovery factor — low	35 %	ASSUMED	Review brief	—
L-05	Recovery factor — mid	37.5 %	DERIVED	(L-04 + L-06) / 2	—
L-06	Recovery factor — high	40 %	ASSUMED	Review brief	—
L-07	Recoverable — low	121.5 MMbbl	DERIVED	L-01 × L-04	—
L-08	Recoverable — mid	139.7 MMbbl	DERIVED	L-02 × L-05	—
L-09	Recoverable — high	159.2 MMbbl	DERIVED	L-03 × L-06	—
L-10	Plateau oil — low	30,000 bopd	ASSUMED	Review brief	—
L-11	Plateau oil — mid	33,507 bopd	DERIVED	(L-10 + L-12) / 2	—
L-12	Plateau oil — high	37,014 bopd	ASSUMED	Review brief	—
L-13	Recoverable gas	790.9 bcf	VERIFIED	Public tracker	S-01
L-14	Gas nameplate	150 MMscfd	VERIFIED	Operator release	S-02
L-15	Oil nameplate	40,000 bopd	VERIFIED	Operator release	S-02
L-16	Water depth	900–1,200 m	VERIFIED	Operator release	S-02
L-17	Location	5.888°N 114.510°E	VERIFIED	Public tracker	S-01
L-18	Working interest	100 % PCSB	VERIFIED	Trade press	S-03
L-19	First oil date	—	OPEN	No public schedule	—
L-20	Economic limit	—	OPEN	Needs fiscal engine	—
L-21	Injection duty	—	OPEN	Basis unresolved	—
L-22	Formation volume factor	1.25 rb/stb	ASSUMED	HERMES — analogue	—
L-23	Plateau share of ultimate	70 %	ASSUMED	HERMES — convention	—
L-24	Boe conversion	5.8 Mcf/bbl	ASSUMED	HERMES — convention	—
L-25	Fiscal terms	—	OPEN	PSC text not public	—

Table B.1 — Assumptions ledger. Every figure in Parts 1 and 3 resolves to a row here. Rows marked OPEN carry no value by design.

Ledger arithmetic

Every DERIVED row can be repeated without trusting this document.

```
L-02 = (347+398)/2 = 372.5
L-05 = (0.35+0.40)/2 = 0.375
L-07 = 347 × 0.35 = 121.45
L-08 = 372.5 × 0.375 = 139.6875
Gas = 150 × 365 / 1000 = 54.75 bcf/yr
790.9 / 54.75 = 14.45 yr
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L-09 = 398 × 0.40 = 159.2
L-11 = (30000+37014)/2 = 33507
Plateau_yr = R × 10⁶ / (rate × 365)
→ 121.45e6/(30000×365) = 11.09
→ 139.6875e6/(33507×365) = 11.42
→ 159.2e6/(37014×365) = 11.78
```

Appendix E — Source register

Only sources cited in this pack are listed.

ID	Source	Accessed
S-01	Global Energy Monitor, **Limbayong Gas Field (Malaysia)**, Global Oil & Gas Extraction Tracker, data year 2023. Field statistics, coordinates, design capacity.	19 Oct 2026
S-02	PETRONAS, **PETRONAS Awards Limbayong Deepwater EPCIC Contract to TechnipFMC**, media release, 1 June 2021. Water depth, distance offshore, well count, facility description.	19 Oct 2026
S-03	Welligence Energy Analytics, **Malaysia FID alert — Limbayong**, 31 August 2026. Final investment decision, working interest, discovery history.	19 Oct 2026
S-04	SPE 210079-MS, **Limbayong: Decoding Industry Top Decile Reservoir Complexity for Marginal Deepwater Development**, 2022. Structure dimensions, complexity characterisation.	19 Oct 2026
S-05	SPE 210549-MS, deepwater drilling margin and managed-pressure-drilling requirement. Referenced in Part 3 context only.	19 Oct 2026
S-06	Upstream, **Petronas finally takes FID on Limbayong deepwater development**, 28 August 2026. FID date and facility concept.	19 Oct 2026
S-07	Dagang NeXchange Berhad, Bursa Malaysia filing, 20 August 2026. Disposal of FPSO **Excalibur** to MISC. Not used in this pack; cited in Part 4.	19 Oct 2026
S-08	Natural Earth, 1:10m coastline, physical vector dataset. Map base for Figure 0. Public domain.	19 Oct 2026

Table E.1 — Source register. Every VERIFIED row in Appendix B resolves to an entry here. Sources are public and independently checkable.

Source limitations

S-01 is a secondary tracker. It reports field statistics compiled by a non-commercial monitor, not by the operator. Its reserve figures are treated as the best available public estimate and not as an operator statement.

S-04 and S-05 are conference papers. They carry the operator's own technical characterisation of the field, which is why they are given weight — but they describe the reservoir, not the development plan, and they do not disclose volumes, costs or schedule.

No source in this register is a primary operator document for volumes, costs or fiscal terms. That absence is the central limitation of the entire pack, and it is the reason so much of this document is OPEN.

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